

Lecture 9 — Descent and Gravity

Keeper, for Casey Koons and the team

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The question

How does a ten-real-dimensional object with no time become the four-dimensional spacetime we live in — and what does it say about gravity?

The object of Lecture 1 is not a spacetime. It has five complex dimensions, a positive-definite metric, and a time that is a rotation of its centre (Lecture 4), not a coordinate. Yet the physics read off it is supposed to be *our* physics, in three space dimensions and one time. This lecture is about the step down, and about what the program can and cannot say once it has taken it. It is the lecture where the word *induced* does the most work, and where we are most careful not to let a relation be called a prediction.

For the reader in a hurry

The shape does not contain our spacetime; it contains a *way down* to it. There is exactly one kind of step — a projection you can take once and not twice — that drops one dimension and leaves three directions of one sign and one of the other. That much is a theorem. *Which* way down is taken is not in the shape: something has to be looking, and the shape cannot say what. So spacetime is induced, not predicted, and we have a test for whether the looking is being done consistently. About gravity we say one clean thing: Newton’s constant and the electron’s mass are tied together by the shape to a tenth of a percent. That is a trade of one measured number for another, not a number from nothing, and we refuse to call it more. The cosmological constant’s *shape* comes out as a balance; its *value* waits on one power we cannot yet fix.

The descent: structure derived, frame input

Start from the isometry group $SO(5, 2)$ and ask what a Lorentzian four-space inside the object would have to be. The answer is constrained by the Jordan structure. The step down is a single idempotent: a projection P with $P^2 = P$, applied once — there is no second step, because applying it again does nothing. It has codimension one — it drops exactly one dimension. And the signature of what it leaves is $(3, 1)$: three directions of one sign, one of the other. In root terms the 3 is the multiplicity of the short roots of B_2 (the multiplicity a of Lecture 2) and the 1 that of the long roots (Lyra, Round 142; an earlier draft said “read off the long root,” which is not T2545’s argument). The three spatial directions are the middle Peirce space of Lecture 2 — $V_{12} \cong \mathbb{C}^3$, the complexified $SO(3)$ vector, irreducible over $\mathbb{C}$ by Jordan structure theory (T2545) — and the one time direction is the centre’s.

That is the structure, and it is derived. One repair rides with it: T2545’s argument for the $(3, 1)$ count was first written from a real-space premise; the conclusion is unchanged, and the reason is now the irreducibility of V_{12} over $\mathbb{C}$, with “real $SO(3)$ vector” naming its Frobenius–Schur type (Grace’s fix, Round 142 G1, commit 704d402d, 2026-09-11; Cal Section 950 scoped the FS clause so that it describes the gap rather than adding a second wall).

Then the honest half. The structure says *what kind* of step down exists; it does not say *which* four-space is selected among the family the step allows. We proved that it cannot: geometry alone does not select the frame (T2565). The selection needs a matter or observer input. So the descent is *Machian, forced-except-one-input* — the same shape of statement as the colour identification of Lecture 2 and the n_C tiebreaker of Lecture 5. We used to write

that the descent was posited; then that it was derived; the correct word is induced: structure from the geometry, selection from what is looking.

What that honesty buys is a falsifier in *shape*, Cal's: $\varepsilon = 0$ frame agreement — if the descent is induced consistently, the frame picked out by radiation and the frame picked out by matter must coincide, and an off-slice baseline $\varepsilon \neq 0$ refutes the picture. For three weeks ε had a name, a value and no host: three corpus sentences carried it and none put it in a formula (Lyra, Round 143 — the finding came first). It has one now, ruled on the rows as written (Cal, Section 957). After the descent a rest frame is a unit timelike vector — a point of $SO_0(3, 1)/SO(3)$, exactly the velocity T2565 says the geometry cannot supply. Radiation's frame is where the photon flux has no dipole; matter's frame is where distant source counts have no dipole, read through Ellis–Baldwin's $D = [2 + x(1 + \alpha)]\beta$. The program's ε is the rapidity between those two rest frames, each read as the $\ell = 1$ moment of a sky scalar on the induced slice; $\varepsilon = 0$ means one boost, not two. That is the only reading P3 and T2565 license. The observable is the Ellis–Baldwin test (1984): the dipole of distant matter — quasars, radio sources — against the CMB dipole. The primary, pinned from the PDF: Secrest et al., *ApJL* 937 (2022) L31 find amplitudes about $3\times$ and $2\times$ the kinematic expectation and directions 45° and 26° off the CMB dipole — outside the 95% positional uncertainties — a joint 5.1σ against “purely kinematic at 370 km/s,” and the authors' own conclusion is intrinsic anisotropy, 48° away. (An earlier draft of this paragraph said “directions agreeing”; it was wrong, from a brief rather than the paper.) Under the ruled reading the measurement is not decidable: reading the matter frame off the dipole presupposes the dipole is purely kinematic, the paper itself rejects that, and that rejection is precisely the escape hatch Cal's pre-registration of 2026-08-21 named before the data. The amplitude excess *would* fire at 5.1σ — on a reading no row makes; it falsifies the cosmological principle, which is Secrest's title, not our row. The decider is a different measurement: the redshift dependence of the dipole. So the register carries ε as a live row in A1's shape — observable, current state, presupposition on its face, kill condition a confirmed kinematic mismatch surviving the hatch (A9, registered 2026-09-14 — falsifier register v0.8; ruling Cal Section 957). An earlier draft called this “the one genuine laboratory-shaped falsifier in this lecture”; now it is one, and it is not yet fired.

Where that stands as this is written (2026-09-14, 11:26). Prior art, found the day the test was registered and not after: Wu & Xia, “Measuring peculiar velocity and tomographic redshift dipole with DESI DR1 catalogs,” arXiv:2608.30914 (31 Aug 2026), measure the boost by the *redshift-dipole* method — the Doppler modulation of observed redshifts — and report for the high-redshift quasar sample $v = 357.95^{+55.05}_{-48.47}$ km/s, in agreement with the CMB-inferred 369.82 ± 0.11 . That is P1's Landing A on a different catalogue by a different estimator, two weeks before our file; so the test is registered as a replication with an independent estimator, not a first test (K1898). The pre-registration (Cal; frozen v1.2 as this paragraph was first written, SHA256 6f036bdb..., gates K1899/K1901; the file the run was certified against is v1.5.2, SHA256 8013d959..., K1909 — the lineage is in the landing paragraph below) fixes, before any catalogue is opened, the catalogue (Quaia; Storey-Fisher et al. 2024, *ApJ* 964, 69), the mask, the bins, the per-bin factor $f_i = 2 + x_i(1 + \alpha_i) + B_i$ with von Hausegger–Dalang's observed-redshift boundary term, and a three-way partition of every outcome written as code — A one boost, B a discrepancy that survives the hatch (the row fires), C no decision at this depth. The two channels are the two factors of a record (Lecture 4, Theorem B): the count dipole is the Rac factor's $\ell = 1$ moment, the redshift dipole the clock factor's. The one thing the synthetic controls taught before the sky was opened is that you must *window one factor and read the other*: a sample windowed on observed redshift and then read on redshift cancels its own boost at first order — the window moves with what it reads, the same mechanism as B_i — so the redshift channel is windowed on flux and the two channels are fitted for one shared boost (Elie 5759; K1900; Lyra R148). At Quaia's depth the count channel alone cannot decide (its synthetic $\sigma \approx 443$ km/s against the pre-registered 185 km/s bar); the joint fit's expected power, computed before its own run, is near 160 km/s (Elie 5760, pre-registration). No catalogue had been opened as this paragraph was written. It will be rewritten and dated when the landing is certified, whichever letter it is. *Landing, certified 2026-09-17 10:12 (K1910, Cal §976, K1911)*: Run 1 on Quaia landed C, not decidable, on both samples. The catalogue was opened on 2026-09-14 and the pre-registration was re-frozen seven times before any dipole was measured, five of them because a synthetic sky carrying the true boost failed the text as written — once because the frozen partition would have called a perfect confirmation a discrepancy; once because v1.5 had omitted a coefficient the frozen per-bin table determines (an instrument error, corrected in v1.5.1, not a prior); and once because the corrected model lost its power, which v1.5.2 restored by writing on the page what every Ellis–Baldwin test in the field assumes silently, a delta prior at zero on the sky's own clustering dipole, and replacing it with a declared zero-mean isotropic Gaussian at ten times the Λ CDM expectation (shape forced by isotropy; only the width a ruling). The sky was then read with the CMB target withheld. Before the

target was consulted, the design's own checks fired: the profile-alternative fit moved the boost by 7.5σ on the primary sample against a frozen bar of 2σ , and the per-bin residuals failed at $p = 0.000$; the per-bin dipoles rise with redshift, which neither a boost nor a clustering dipole does, and which the whole-sample selection map used as a fallback weighting reproduces on synthetic skies (Elie 5767) — a selection systematic, with a second layer the fallback does not explain. The fitted vector on the primary sample, read afterwards, is consistent with the CMB boost (28° off, $\chi^2_3 = 3.7$) and preferred over no boost — but a vector from a model that does not describe the sky is not evidence either way, and the text says so. A9 stays live. Quia at this weighting is not a test of P1; the next catalogue is chosen on that, and nothing runs before a fresh freeze.

Gravity as a relation, not a prediction

The program's gravity result is one equation,

$$G = \hbar c \frac{(6\pi^5)^2 \alpha^{24}}{m_e^2},$$

which reproduces Newton's constant to 0.065% from the electron mass. For a while we called this “Newton's constant predicted.” We do not now, and the reason is exactly the reason a referee would give.

The relation is a theorem (the paper “Gravity: the 2→1 Reduction,” Keeper pass K1673): the geometry ties the gravitational scale to the electron's, with the exponent $24 = 4C_2$ as a supporting identity (Cal Section 585) — the Bergman round-trip doubled by $G \propto m_{\text{Planck}}^{-2}$. But every dimensionful prediction needs a dimensionful input; no theory produces a G with no scale in it. What the relation does is trade one dimensionful input for another: take m_e as the ruler and G follows, or take G and m_e follows. It reduces two inputs to one. And it carries α to the twenty-fourth power, and α is identified, not derived (Lecture 8) — so the relation's *precision* rests on a measured α as well.

Two further honesties. Six historical readings of this same relation appear across the corpus (F66, T201, T1918, T1955, T1301a, the paper); they count as one result, not six confirmations. And a Kaluza–Klein route to the same number, once cited as independent, was found circular — it fed ℓ_B back into itself — and is demoted (K1408).

So the sentence is: *the geometry ties G to m_e , and we take m_e as the ruler*. Not: *the geometry predicts G* .

The foundation for a gauge theory — and what it is not

Can a Yang–Mills theory live on the object? The foundation questions were settled in August, and the settlement is worth stating because the legacy chapters overstated it.

The physical Hilbert space is $H^2(D_{IV}^5)$, on which $G = \text{SO}_0(5, 2)$ acts irreducibly and unitarily (a scalar-type highest-weight representation at $\nu = 5/2$; it is the Bergman space, at $\nu = 5$, that is holomorphic discrete series — Lyra's pin, Round 142; the W1 cell's label is flagged); that the group acts was the lever that decided the question between the two spaces (W1). On H^2 directly, locality and the modular structure carry over *verbatim*: the Tomita–Takesaki modular theory, the Bisognano–Wichmann relation between the modular group and boosts, and the transport of locality along it — standard theorems, cited and not claimed. The Szegő projection onto the boundary is exact (T349). So the foundation clears — this is the cell the program calls G3.

It is *not* the Clay problem. Whether the resulting theory is genuinely interacting or a generalised free field — G6 — is open, and the mass gap is not claimed. A legacy Guide chapter once said otherwise; it is corrected in place. And one demotion that is not a dropping: the arithmetic quotient $\Gamma(137) \backslash D_{IV}^5$, which once carried the QFT lane, is retained only as the *arithmetic-lane* regulator (the Selberg trace formula that produced the a_e match of Lecture 8); the QFT lane runs on H^2 (Cal R46, K1785).

Λ : a closed form with one obstacle

The cosmological constant is 10^{120} times smaller than a naive estimate, and any theory of this kind is asked about it. The program's answer improved in August from “a structural floor” to a closed structural form with one named

obstacle.

The form is a thermostat: a fixed-point balance in which the Koons tick t_K (Lecture 4) is load-bearing,

$$\frac{\Lambda}{\Lambda_P} = \left(\frac{t_K}{t_P} \right)^{2p/(2-p)},$$

with t_P the Planck time and p a *mismatch power* — the $5 \rightarrow 4$ reduction of the residual. The balance exists; both controls pass (one of them, T2571, reproduced *un-fed*, which is what made the control meaningful); the response coefficient enters from the equation’s own structure, and a “curvature quantum” that an earlier version needed was shown unnecessary. The one free thing is p . Its domain is $(0, 2)$; the geometry has not fixed it; and until it does, no evaluation of Λ exists and none is claimed. The form is capped *conditional* at every site where it is quoted (K1057’s lineage), and the quarantine on the value is forward-transferred: whoever forces p computes the number blind. A forced p outside $(0, 2)$ falsifies the form.

Note, 2026-09-22 (K1919 §6–7; Casey’s definition of the tick as the time to begin a commitment). The one free thing is no longer free: the Peirce weight of the spatial sector fixes $p = 1$ (Lyra, ITEM6; the kill written first), and at $p = 1$ the exponent is 2, under which the thermostat reads $\Lambda/\Lambda_P = (t_K/t_P)^2$. With t_K the begin-time of a commitment at the horizon temperature, $t_K/t_P \simeq H t_P$ and the form is de Sitter’s $\Lambda \propto H^2$ — Friedmann’s $\Lambda = 3H^2 \Omega_\Lambda$ to the digit (CODATA, Planck 2018: $(H_0 t_P)^2 = 1.4 \times 10^{-122}$ against $\Lambda/\Lambda_P = 2.9 \times 10^{-122}$, ratio $2.08 = 3\Omega_\Lambda$). So the form is a *consistency with GR*, not a derivation of Λ ’s value: H is measured because it changes, and no tick in the corpus produces it from the five integers. The obstacle moved: the open object is the ledger’s *filling law* — how the committed count’s fraction of the horizon count changes with the scale factor — which fixes $w(z)$ and is pre-registrable against DESI DR2. The earlier tick, $t_P \alpha^{36}$ (T2405), was fitted to a number that was never a time and is re-labelled on the registry.*

The wall, named

Could the object carry its own gauge dynamics on the spin-2 sector — an Einstein equation, or an $su(3)$ native to the geometry? Two theorems say no, in the only sense that closes a class (Lane I). Every K -equivariant antisymmetric bilinear on the relevant carrier contains spin-3; and the typed write channel — the one non-equivariant ingredient the program owns — compresses onto the carrier where dimension forbids spin-3. So the write channel explains *why* there is no native $\ell = 2$ dynamics and supplies none. The door is named: break equivariance or antisymmetry, and say which. Separately, the once-hoped identification of the object’s spectral action with its causal order sits at an honest floor (K1674): two candidate weightings failed blind, the tier is “no surviving candidate,” not “impossible,” and the *derived causal order* — the real result — stands.

Tier line

- Derived: the descent’s structure ($P^2 = P$, codimension one, $(3, 1)$); the causal order.
- Input: the frame (T2565).
- Theorem about a relation: $G \leftrightarrow m_e$ at 0.065%, α identified inside.
- Cleared: the gauge-theory foundation on H^2 (G3). Open: G6.
- Floored, doors named: Λ ’s value (p); the $\ell = 2$ native dynamics; the spectral-action weighting.
- Not claimed: G from nothing; Λ derived; the mass gap; an Einstein equation on the object; the descent as predicted.

What would make this lecture wrong

A confirmed kinematic boost mismatch between the radiation and matter rest frames that survives the pre-registered hatch — the redshift dependence of the matter dipole is the decider (A9). A forced p outside $(0, 2)$. A G – m_e relation off 0.065% once α ’s running is accounted at the stated scale. A native spin-2 dynamics exhibited without breaking equivariance or antisymmetry — which the theorems say cannot happen, so an exhibit would be a discovery about the theorems.

Where to look

The descent row (rubric, 2026-08-22) and T2565; T2543/T2545 with the Round 141 reason-fix; “Gravity: the $2 \rightarrow 1$ Reduction” with K1673, K1408 and Cal Section 585; the YM foundation cell (2026-08-22) and T349; Lane Λ (2026-08-26) with T2571 and the K1057 cap; Lane I; K1674.